



Learned bounded refinement
Top- k support + LS
Local scalar or CDL
feature controller
(distinct validated paths)

Channel NMSE

Delay + projected-angle
estimates

Sparse angle-delay-
Doppler observation

Hungarian matching + alias-aware physical loss

HELD-OUT EVIDENCE

+6.23 dB

CDL-A/C channel NMSE

-10.12 ns

CDL-A/C delay RMSE

Zero-shot CDL-D

exposes profile-shift limit

Interpretable bounded refinement around selected tensor atoms; failure boundaries kept visible.